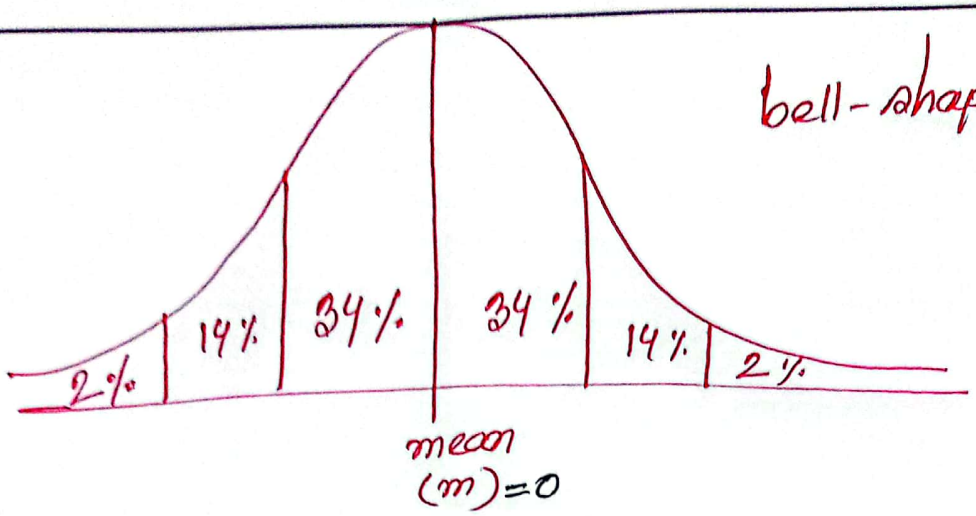
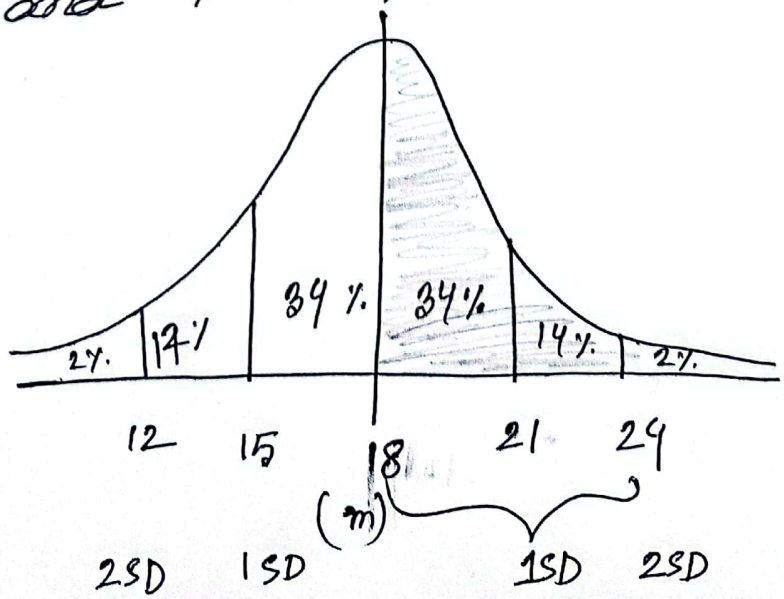
Normal Distribution



Q.1. The hourly wage paid to working adults in Maplewood is normally distributed around a mean of \$18 per hour with a standard deviation \$3. Find the percent of working adults who are paid between \$18 and \$24 per hour, inclusive, draw the curve.

Sol^n:



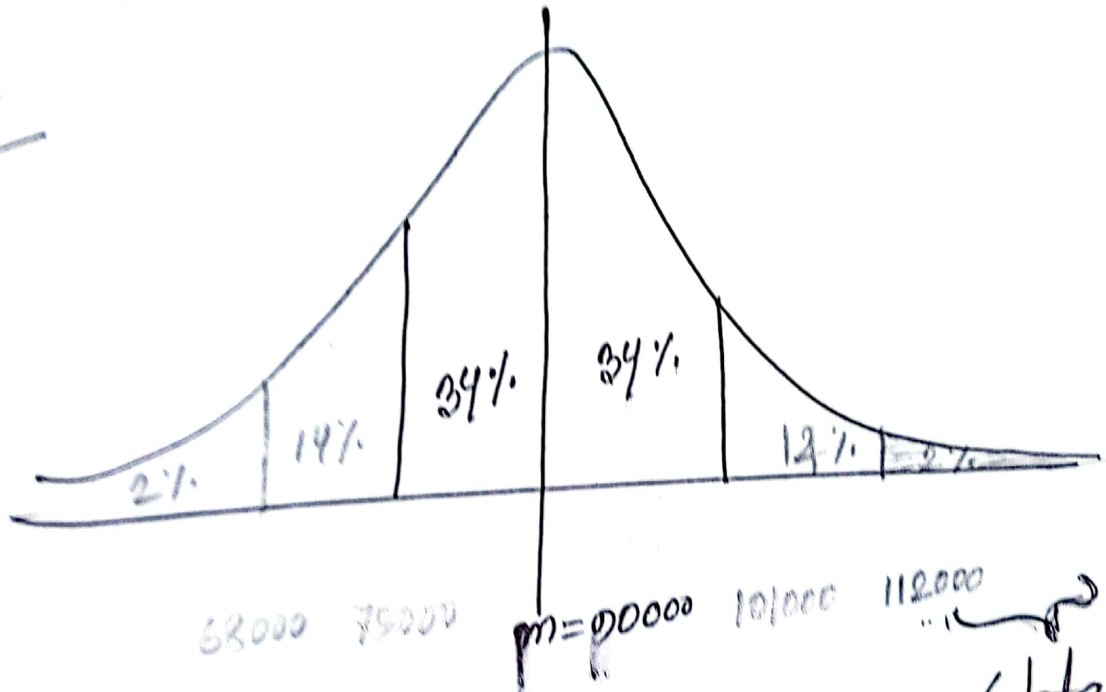
$(34+14)\%$

$= 48\%$

(ans)

Q.2 Home values among the 8000 homeowners of Town X are normally distributed, with a standard deviation of \$ 11,000 and a mean of \$90,000. Determine the number of homeowners whose home value is above \$ 112,000.

Soln:



∴ Numbers of homeowners

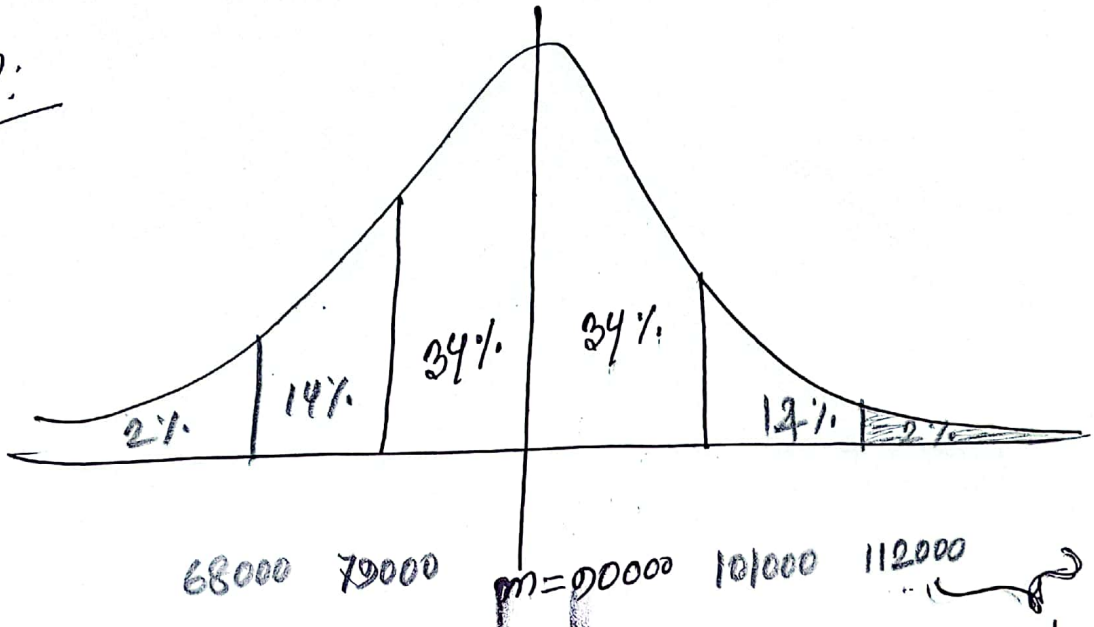
2% of total homeowners

$= \frac{2}{100} \times 8000$

$= 160$ (ans).

Q.2. Home values among the 8000 homeowners of Town X are normally distributed, with a standard deviation of \$ 11,000 and a mean of \$90,000. Determine the number of homeowners whose home value is above \$ 112,000.

Soln:



(data stand at this plot).

∴ Numbers of homeowners

2% of total homeowners

$= \frac{2}{100} \times 8000$

$= 160$ (ans).